

In the name of God the compassionate the merciful

Title:

ChatGPT in RStudio with gptstudio - Better than
GitHub Copilot?

Speaker:

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The Objective of This Event

The purpose of this event is to demonstrate how large language models (LLMs) such as GPT-5, GPT-4 and GPT-3.5-Turbo (commonly known as ChatGPT) can significantly enhance the R programming workflow.

With the introduction of the new gptstudio package in R, users can seamlessly integrate AI-powered assistants directly into RStudio, enabling features such as code suggestions, automated documentation, and intelligent debugging.



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How to Add ChatGPT to Rstudio?

Install the **gptstudio** package

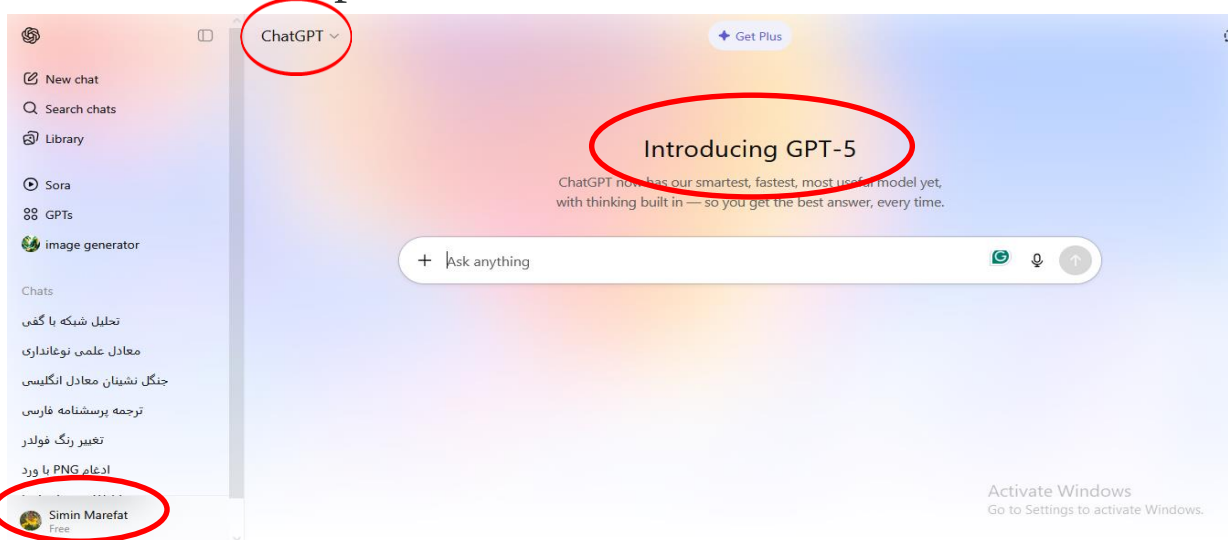
Load the required packages:

```
install.packages('pacman')  
library(pacman)  
pacman::p_load(devtools, usethis)  
pacman::p_load_gh("MichelNivard/gptstudio")
```



Make an OpenAI account

You need an OpenAI account to interact with ChatGPT:



Create an OpenAI API key

To use the Open AI package within Rstudio, you need an OpenAI API Key:





Create

○ Chat

🔊 Audio

🖼️ Images

🗣️ Assistants

Manage

🔑 Keys

🔑 API keys

📁 Logs

📁 Storage

📁 Batches

🔧 Optimize

🔍 Evaluations

⚙️ Fine-tuning



API keys

You have permission to view and manage all API keys in this project.

Do not share your API key with others or expose it in the browser or other client-side code. To protect your account's security, OpenAI may automatically disable any API key that has leaked publicly.

View usage per API key on the [Usage page](#).

+ Create new secret key



Create an API key to access the OpenAI API

+ Create new secret key

Save your key

Please save your secret key in a safe place since **you won't be able to view it again**. Keep it secure, as anyone with your API key can make requests on your behalf. If you do lose it, you'll need to generate a new one.

[Learn more about API key best practices](#)

sk-proj-OyeFYaogPOlof2wVS2GdSr_MkE

Copy

Permissions

Read and write API resources

Done



Copy the secret key
and store it
somewhere safe,
then click Done.

Set the API key in Rstudio

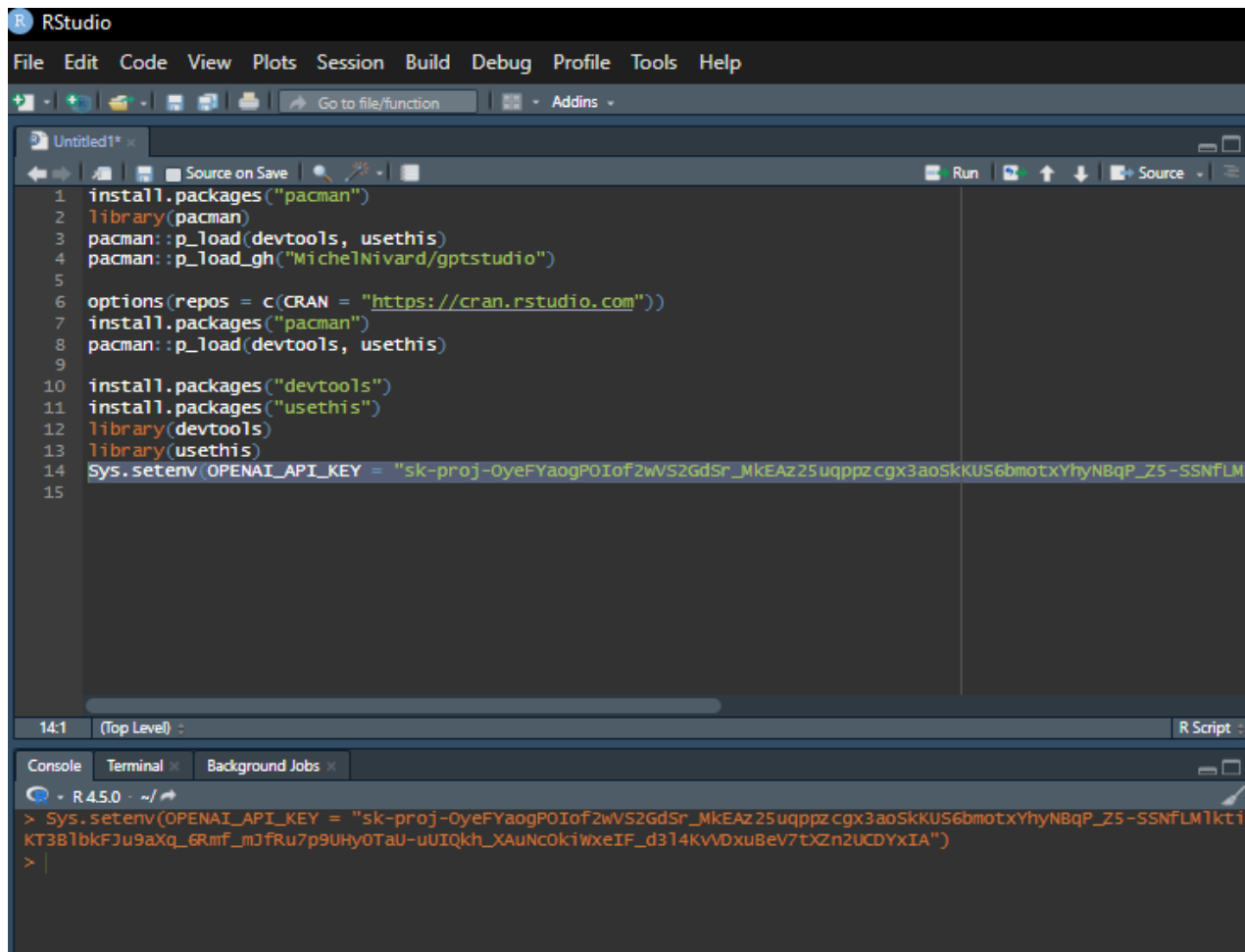
For RStudio to connect to ChatGPT, the OpenAI API key must be set within Rstudio:

Notic: To set the API key you can either:

```
Sys.setenv(OPENAI_API_KEY = "<APIKEY>")
```

```
Sys.setenv(OPENAI_API_KEY = "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")
```





```
1 install.packages("pacman")
2 library(pacman)
3 pacman::p_load(devtools, usethis)
4 pacman::p_load_gh("MichelNivard/gptstudio")
5
6 options(repos = c(CRAN = "https://cran.rstudio.com"))
7 install.packages("pacman")
8 pacman::p_load(devtools, usethis)
9
10 install.packages("devtools")
11 install.packages("usethis")
12 library(devtools)
13 library(usethis)
14 Sys.setenv(OPENAI_API_KEY = "sk-proj-0yeFYaogPOIof2wVS2GdSr_MkEAz25uqppzcgx3aoSkKUS6bmotxYhyNBqP_Z5-SSNfLM1kt1
15 KT3B1bkF3u9aXq_6Rmf_mJfRu7p9UHy0TaU-uUIQkh_XAuNcokiWxeIF_d314KvVDxuBeV7tXZn2UCDYxIA")
16
```

14:1 (Top Level) R Script

Console Terminal Background Jobs

R 4.5.0 ~ /

```
> Sys.setenv(OPENAI_API_KEY = "sk-proj-0yeFYaogPOIof2wVS2GdSr_MkEAz25uqppzcgx3aoSkKUS6bmotxYhyNBqP_Z5-SSNfLM1kt1
KT3B1bkF3u9aXq_6Rmf_mJfRu7p9UHy0TaU-uUIQkh_XAuNcokiWxeIF_d314KvVDxuBeV7tXZn2UCDYxIA")
>
```



```
usethis::edit_r_environ()
```

```
usethis::edit_r_environ()
```

Add the following line to the file, replacing "<API KEY>" with your actual secret key:

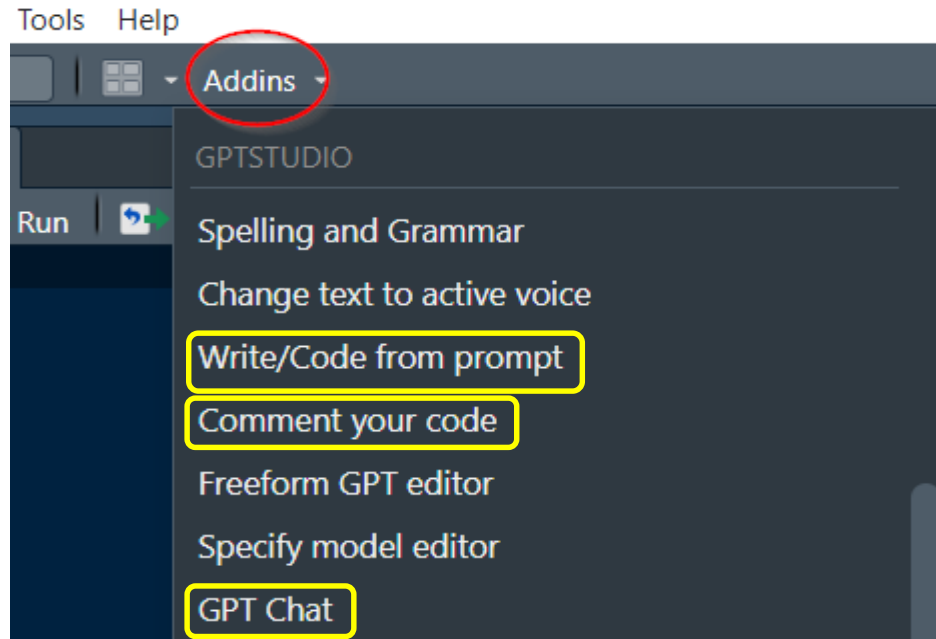
```
OPENAI_API_KEY= sk-proj-  
OyeFYaogPOIof2wVS2GdSr_MkEAz25uqppzcgx3aoSkKUS6bmotxYhyNBqP_Z5-  
SSNfLMlktiKT3BlbkFJu9aXq_6Rmf_mJfRu7p9UHy0TaU-  
uUIQkh_XAuNcOkiWxeIF_d3l4KvVDxuBeV7tXZn2UCDYxIA
```

This API key is now accessible in all your projects which means ChatGPT will work in all your projects.



How to Add ChatGPT to Rstudio?

The **gptstudio** options can be accessed through the **RStudio Addins** menu



Ctrl+Shift+P

Command+Shift+P

GPT Chat

Selecting GPT Chat allows you to interact with ChatGPT directly within Rstudio.

It feels similar to the web app version of ChatGPT and means you don't have to login to ChatGPT with a web browser.



Write Prompt

Chat

Model Input Settings

OpenAI Model

text-davinci-003

Temperature

0.5

Max Tokens

200

Chat History

Question
What is chatGPT?

Response
ChatGPT is a natural language processing (NLP) model developed by Microsoft that is designed to generate human-like conversations. It is based on the GPT-3 model, which is a powerful language model used for natural language processing tasks. ChatGPT is capable of understanding context and responding to questions in a human-like manner.



Write/Code from Prompt

The Write/Code from Prompt option takes a chunk of highlighted text and attempts to write code.

For example, I gave it this text: Use ggplot2 to make a scatter plot of mpg vs. hp using the mtcars dataset.

And it wrote the following code:

```
library(ggplot2)

ggplot(data = mtcars, aes(x = hp, y = mpg)) +
  geom_point()
```



Comment your Code

The Comment your code option takes a selected code chunk and returns the same code with added comments.

For example, I gave it this code chunk:

```
library(ggplot2)

ggplot(data = mtcars, aes(x = hp, y = mpg)) +
  geom_point()
```

And it returned this:

```
library(ggplot2) #Loads the ggplot2 package

ggplot(data = mtcars, aes(x = hp, y = mpg)) + #creates a scatterplot
of mpg vs hp
  geom_point() #adds points to the scatterplot
```



GitHub Copilot vs. gptstudio



Final Thoughts

ChatGPT is not only fun, but is incredibly useful when developing code.

I thought adding ChatGPT to RStudio would be a bit of a gimmick.

But boy I was wrong!

gptstudio brings the power of ChatGPT into RStudio and adds some impressive functionality.



Any Questions?



Thanks for your attention



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